A Computational Companion to Pavliotis, Chapter 1

Master Plan: Covariance Operators, Spectral Theory, and the Karhunen–Loève Expansion

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July 24, 2026

Abstract

This is the *master plan* for a computational project in Julia that deepens understanding of Chapter 1 of Pavliotis, *Stochastic Processes and Applications*, by implementing its core objects and verifying each against an analytic prediction. It owns the project's through-line, its decomposition into feature briefs, the repository architecture and its limits, the cross-cutting conventions, a consolidated risk register, and the budget. It deliberately does *not* carry call signatures, code, or numerical tolerances: those are resolved downstream — in the per-commit plans that each feature brief is handed to, and in the shipped code itself — and a copy here would be a second, drifting source of truth. The project is organized around a single hub, the covariance operator $\mathcal{C}$, with two diagonalizing bases (Fourier/Bochner and Mercer/Karhunen–Loève) and *four* mathematical square roots used for sampling (Cholesky, KL truncation, random Fourier series, circulant embedding) — of which *three are implemented* (Cholesky, KL, circulant); the random-Fourier series is described as structure but not built, and this costs no capability because circulant embedding is itself the Bochner square root. Ergodicity closes the loop between one simulated path and the analytic objects it is checked against. The plan excludes machine-learning and SciML content. At the working budget of **15 h/week** ($6 \times 2.5\,\mathrm{h}$) the full planned ladder is **Units 0–7**: the covariance/spectral spine (Units 0–4), Brownian motion as a scaling limit (Unit 5), fractional Brownian motion (Unit 6), and a forward-pointing SDE bridge to Chapter 3 (Unit 7). **Units 0–3 are shipped**; Units 4–5 are committed core; Unit 6 then Unit 7 are the cut order if the buffer is consumed. Each unit is written as a feature brief — its one idea, the claim it proves and the exact analytic target it is checked against, the falsifier it is designed to exhibit, its place in the build DAG, the repository areas it touches, its effort and dominant risk, and its status. Statements drawn from the text are cited by section/theorem number; statements standard but external to Pavliotis are flagged explicitly. All citations are to the project's copy of the text, the *November 11, 2015 draft* (numbering differs from the published edition), checked directly against it.

Contents

1 The Through-Line

1.1 The hub: the covariance operator

Every object studied in Chapter 1 is a fact about a single operator. For a mean-zero, second-order process X_t, $t \in [0,T]$, with continuous covariance $R(t,s) = \mathbb{E}(X_t X_s)$, define the integral operator

$$(\mathcal{C}f)(t) \;:=\; \int_0^T R(t,s)\, f(s)\, ds, \qquad f \in L^2[0,T]. \tag{1}$$

(Pavliotis, eqn. 1.33) This is self-adjoint, nonnegative, and (for continuous R) compact and Hilbert–Schmidt (Pavliotis, Exercises 21–22). Everything downstream is one of two operations performed on $\mathcal{C}$.

1.2 Two diagonalizations

Fourier / Bochner (gated by stationarity). If X_t is weakly stationary, $R(t,s) = R(t-s)$ is a function of a single lag, and Bochner's theorem applies:

$$R(\tau) \;=\; \int_{\mathbb{R}} e^{i\omega\tau}\, \rho(d\omega), \tag{2}$$

for a unique nonnegative measure ρ on $\mathbb{R}$ with $\rho(\mathbb{R}) = R(0)$ (Pavliotis, Thm. 1.14). When ρ is absolutely continuous, $\rho(d\omega) = S(\omega)\, d\omega$ with

$$S(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\omega\tau} R(\tau)\, d\tau, \qquad R(\tau) = \int_{-\infty}^{\infty} e^{i\omega\tau} S(\omega)\, d\omega. \tag{3}$$

(Pavliotis, eqns. 1.7–1.8) $S(\omega)$ is the *spectral density*; it lives on the frequency axis, not on the sample space — a distinction worth stating explicitly since both objects are conventionally denoted ω. Equations (3) form a Fourier transform pair and invert by the Fourier inversion theorem, independently of any symmetry of R. Realness and evenness of R add only that S is itself real and even — so the sign of i in the exponent is immaterial (one may replace $e^{\pm i\omega\tau}$ by $\cos\omega\tau$); the $1/2\pi$ is placed on the S-side purely by convention. With this convention $\int_{\mathbb{R}} S(\omega)\,d\omega = R(0)$ (total variance), which fixes the normalization every estimator in Unit 1 must reproduce.

Mercer / Karhunen–Loève (general, needs a compact domain). Independently of stationarity, the spectral theorem for compact self-adjoint operators gives a countable orthonormal eigenbasis $\{e_k\}$ of $\mathcal{C}$ with eigenvalues $\lambda_k \geq 0$,

$$\mathcal{C}e_k = \lambda_k e_k, \qquad R(t,s) = \sum_{k=1}^{\infty} \lambda_k\, e_k(t)\, e_k(s), \tag{4}$$

(Pavliotis, eqn. 1.32; Mercer's theorem, stated inline (unnumbered) in §1.5) and the Karhunen–Loève expansion

$$X_t = \sum_{k=1}^{\infty} \xi_k\, e_k(t), \qquad \xi_k = \int_0^T X_t\, e_k(t)\, dt, \quad \mathbb{E}(\xi_k \xi_\ell) = \lambda_k \delta_{k\ell}, \tag{5}$$

(Pavliotis, Thm. 1.28 (Karhunen–Loève)) converging in L^2, uniformly in t. If X_t is Gaussian, the ξ_k are independent (uncorrelated *and* jointly Gaussian), and

$$X_t = \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, \xi_k\, e_k(t), \qquad \xi_k \overset{\text{iid}}{\sim} N(0,1). \tag{6}$$

(Pavliotis, eqn. 1.36) The eigenvalues are simultaneously (i) eigenvalues of the covariance operator and (ii) variances of the independent stochastic coefficients in the expansion — one fact, two readings.

When the two diagonalizations coincide. The Fourier basis diagonalizes $\mathcal{C}$ exactly when $\mathcal{C}$ is a convolution operator, which requires the domain to be a group under addition and the kernel to be translation-invariant. The interval $[0,T]$ is not a group; the circle (torus) is. On the torus, $\mathcal{C}$ is a genuine convolution, its eigenfunctions are forced to be the characters e^{ikt}, and

$$\lambda_k = \hat{R}(k) \qquad \text{(exactly)}, \tag{7}$$

the k-th eigenvalue equals the k-th Fourier coefficient of R — the KL sequence and the (discrete, atomic) spectral measure are the same object. On $[0,T]$ the boundary breaks translation invariance: the KL eigenfunctions of Brownian motion, for instance, are Sturm–Liouville sine modes with eigenvalues $\lambda_n \propto (2n-1)^{-2}$ (Pavliotis, Example 1.29), not Fourier modes, and (7) fails. *[External / not in Pavliotis: the general asymptotic recovery $\lambda_k \to \hat{R}(\omega_k)$ as $T \to \infty$ — where $\hat{R}(\omega) = 2\pi S(\omega)$ is the* un-normalized *operator symbol, since the eigenproblem $\int R(t,s)e(s)\,ds = \lambda e$ carries no $1/2\pi$ — is a Toeplitz-operator fact (Grenander–Szegő), not covered in Pavliotis. Stated precisely, Szegő controls the eigenvalue* distribution *— linear statistics $\frac{1}{n}\sum_k f(\lambda_k)$ converge to the average of $f \circ \hat{R}$ over the frequency band — and* not *each eigenvalue uniformly in k; the extreme eigenvalues, at the edges of the spectrum, are exactly where that control is weakest. This asymptotic — not an exact $[0,T]$ identity, and not a uniform-in-k one — is precisely the object of the Unit-3 cross-check, and the distributional/uniform distinction is why that cross-check is restricted to the resolved bulk of the spectrum rather than gated on a $\max_k$ over all of it.]*

1.3 One factorization, four routes (three implemented)

Sampling a Gaussian process is applying a square root of $\mathcal{C}$ to white noise. Four square roots give four sampling algorithms, all reproducing the same *law*; **three are implemented** (Cholesky, KL truncation, circulant embedding). The random-Fourier series is described here as genuine mathematical structure but is *not built* — and, as noted below, this leaves no capability gap.

- **Cholesky (general; implemented):** $\Sigma = LL^\top$ on a grid, $X = Lz$, $z \sim N(0, I)$.
- **KL truncation (general; implemented):** (6) truncated at K terms.
- **Random Fourier series, Gaussian amplitudes (stationary only; *described, not implemented*):** on a *one-sided* frequency grid $\omega_k > 0$, $X(t) = \sum_k \big[a_k \cos(\omega_k t) + b_k \sin(\omega_k t)\big]$ with $a_k, b_k \overset{\text{iid}}{\sim} N\big(0,\, 2\, S(\omega_k)\, \Delta\omega\big)$. The factor 2 folds the two-sided spectrum onto $\omega > 0$: $\mathrm{Cov}(\tau) = \sum_k 2S(\omega_k)\Delta\omega \cos(\omega_k \tau) \to R(\tau)$. *Gaussian* amplitudes make this a genuine square root of $\mathcal{C}$ applied to white noise, so the finite-K sample is exactly Gaussian; a random-*phase* variant (fixed amplitude $\sqrt{4S(\omega_k)\Delta\omega}$, uniform phase) reproduces the covariance but *not* the Gaussian law at finite K (Gaussian only in the limit, by the CLT), and is therefore not a square root. (This last is a decision record — the reason the random-phase variant is rejected as a sampler — and is retained even though the whole route is unimplemented.)
- **Circulant embedding (stationary only; implemented):** FFT-diagonalization of the circularly-extended Toeplitz covariance (Wood–Chan / Dietrich–Newsam); the circulant eigenvalues are the DFT of the covariance sequence — i.e. the *discrete* spectral density — so this is precisely the Bochner square root made computational, on the grid rather than on the line. Exact and $O(m \log m)$ when the embedding is nonnegative definite.

Cholesky and KL are general; the random-Fourier and circulant routes require stationarity. Each is a different basis in which to take the square root of the same operator: Cholesky is the triangular factor, KL is the eigenbasis, and random-Fourier / circulant embedding are the Bochner square roots on the continuous frequency axis and the discrete FFT grid respectively. **Because those two are the *same* Bochner square root in two realizations, shipping only circulant loses no capability** — the Bochner square root is present; the unbuilt route is the continuous-axis twin of a route that is built.

1.4 Stationarity as gate, ergodicity as loop-closer

Stationarity is a *precondition* for the Fourier vertex: Brownian motion is not stationary (its covariance $\min(t, s)$ is not a function of $t - s$), so it has no spectral density; it possesses only the Mercer/KL vertex. Ergodicity is the mechanism that licenses every theoretical check in this project: for a stationary process with $C \in L^1(0, \infty)$,

$$\lim_{T\to\infty} \mathbb{E}\left|\frac{1}{T}\int_0^T X_s\, ds - \mu\right|^2 = 0, \tag{8}$$

(Pavliotis, Prop. 1.16, eqn. 1.12) with finite-T rate (for $C \geq 0$)

$$\mathbb{E}\left|\frac{1}{T}\int_0^T (X_s - \mu)\, ds\right|^2 = \frac{2}{T^2}\int_0^T (T - u)\, C(u)\, du \;\leq\; \frac{2}{T}\int_0^\infty C(u)\, du \;=\; \frac{2D_\star}{T}, \tag{9}$$

(Pavliotis, eqn. 1.13, Lemma 1.17) where the first equality is general (Lemma 1.17) but the inequality uses $C \geq 0$ (for sign-changing C the honest bound is $\frac{2}{T}\int_0^\infty |C|$, and $D_\star = \int_0^\infty C$ coincides with it only in the sign-definite case; the asymptotic form $\mathrm{Var} \sim 2D_\star/T$ holds regardless of sign given integrability). Here we introduce the **Green–Kubo (transport) coefficient**

$$D_\star := \int_0^\infty C(t)\,dt, \qquad \mathbb{E}\Big(\int_0^t X_s\,ds\Big)^2 \approx 2D_\star t \quad (t \gg 1). \tag{10}$$

(Pavliotis, eqns. 1.14–1.15) **Notation caution (carry α).** Pavliotis denotes both the transport coefficient $\int_0^\infty C$ (eqn. 1.14) *and* the noise strength in the OU kernel via $C(0) = D/\alpha$ (Example 1.15) by the single symbol D. Example 1.18 makes the collision explicit: there Pavliotis computes the diffusion coefficient of the exponential-correlation process as $\int_0^\infty C = C(0)\,\tau_{\mathrm{cor}} = D/\alpha^2$, so for the OU kernel the two quantities differ, $D_\star = \int_0^\infty C = D/\alpha^2$, coinciding only at $\alpha = 1$. (Eqn. 1.11 merely sets $\alpha = D = 1$ for a specific covariance computation, incidentally collapsing the distinction — it is not the locus of the conflation.) We keep D for the noise strength and $D_\star$ for the transport coefficient throughout, so every rate constant carries its explicit α-dependence. This is what justifies estimating a covariance or a spectral density from a *single* simulated path and comparing it to a closed-form target — the core verification methodology used throughout.

1.5 Summary

	Fourier / Bochner	**Mercer / KL**
Requires	weak stationarity	compact domain (continuous $\mathcal{C}$)
Object	spectral density $S(\omega)$	eigenpairs (λ_k, e_k)
Lives on	frequency axis $\mathbb{R}$	discrete index $k \in \mathbb{N}$
Exact coincidence	on the torus only: $\lambda_k = \hat{R}(k)$	

2 Feature Ladder, Spine, and Cut Line

Narrative spine vs. build DAG. The *narrative* spine — the order in which the units are best read — is $0 \to 1 \to 2 \to 3 \to 4$, with Units 5, 6, and 7 hanging off it. That order is pedagogical and is *not* the dependency graph. The *build DAG* — what each unit actually reuses, and therefore what each feature brief's "depends on" field records — is

$$1 \Leftarrow 0, \quad 2 \Leftarrow 0, \quad 3 \Leftarrow 0,1,2, \quad 4 \Leftarrow 0,1, \quad 5 \Leftarrow 0,3, \quad 6 \Leftarrow 1,2,4, \quad 7 \Leftarrow 3.$$

In particular **Unit 2 depends on Unit 0 only, not Unit 1** (the KL solver uses linear algebra, not the spectral machinery); $1 \leftrightarrow 2$ is a narrative adjacency, not a build edge. Two of these edges are *preferential rather than strict*: Unit 4 samples OU via Unit 1's circulant route *preferred*, but Unit 0's Cholesky suffices (OU is well-conditioned); and Unit 7 borrows the OU *kernel* that first appears with Unit 1 and checks against Unit 3's exact stationary OU. No edge points forward, so each unit is plannable against only the units below it.

Committed core and the cut line. Units 0–3 are shipped. Units 4 and 5 are committed core. **Unit 7 is the single cut line** — a labelled forward pointer to Chapter 3 that nothing in Units 0–6 depends on, so it is the first to go if the debugging buffer is consumed; **Unit 6 trims second**. (The source this plan replaces contradicted itself here, marking Unit 7 "the single cut line" while also calling Unit 6 material to "trim first alongside 7"; the reconciliation is deliberate: 7 first because it is a pure forward pointer, 6 second because it is a genuinely new kernel that the core does not require.) This protects depth over breadth: the budget beyond the core is spent on *more of Chapter 1 done thoroughly*, not on rushing ahead. See the Budget for the hour split.

3 Feature Briefs

Each brief is orientation for the planner (`plan-and-dispatch`) that will decompose it into commit plans, not a specification. Its fields: **Idea** (the one concept); **Proves** (the claim and the exact analytic target it is checked against, with any external target flagged); **Falsifier** (the hypothesis whose failure the unit is designed to exhibit — the intent of its negative control); **Depends on / enables** (its place in the build DAG); **Deliverable shape** (the repository areas it touches and the class of artifact it emits, in prose — never a call signature or a tuned number); **Effort & risk** (the estimate and the dominant overrun risk, which points into the risk register); and **Status** on two axes — *build* (shipped / planned) and *cut* (core / trim-second / cut-line-trim-first). The exact closed-form targets below are philosophy and are pinned here; the tolerances that gate them are measurement and live in the code and the per-commit plans.

Unit 0 — Covariance Core and Gaussian Sampling via Cholesky

Idea A Gaussian process is fixed by its mean and covariance (Pavliotis, App. B.5); sampling is applying a square root of Σ (Cholesky, $\Sigma = LL^\top$, $X = Lz$).

Proves The empirical covariance of N sample paths converges to the analytic $\Sigma_{ij} = R(t_i, t_j)$ at the Monte-Carlo rate. The headline artifact is the fitted log–log slope of $\|\widehat{\Sigma}_N - \Sigma\|_F$ against N, checked against the analytic $-\frac{1}{2}$ and gated stochastically (see conventions); the intercept is irrelevant, the exponent is the object. A passing slope certifies the sampler and the estimator at once.

Falsifier The nugget is load-bearing. Factoring the *same* singular Brownian-motion Σ used in the main check — its $t = 0$ row is identically zero, $R(0, s) = 0$, so Σ is exactly singular — with the nugget switched off throws; the throw disappears once the nugget is restored. Reusing the main-check Σ (rather than a separate pathological matrix) is the honest control: the OU kernel is well-conditioned and would not throw.

Depends on / enables Root of the build DAG (depends on nothing). Enables Units 1–5: every later unit draws paths through this.

Deliverable shape `gaussianprocess.jl` (the `GaussianProcess` type, grid assembly of Σ, the empirical-covariance estimator) and `sampling.jl` (the jittered Cholesky route). The `experiments/00_covariance_core` experiment emits a sample-paths figure and the error-vs-N log–log plot with the fitted slope annotated against the reference line, and records the jitter and the RNG seed.

Effort & risk $\approx$ 6–8 h (includes the repository skeleton). Dominant risk: path-matrix orientation — `empirical_cov` expects paths as *columns* ($n_{\text{grid}} \times N$); a transpose silently estimates the wrong matrix.

Status Build: **shipped**. Cut: **core**.

Unit 1 — Stationarity Gate: Spectral Density vs. Bochner (with Circulant Embedding)

Idea Weak stationarity (Pavliotis, §1.2) moves all second-order structure onto the frequency axis (Bochner, (2)); the Ornstein–Uhlenbeck / exponential kernel has a Lorentzian spectrum,

$$R(\tau) = \frac{D}{\alpha} e^{-\alpha|\tau|} \quad \Longrightarrow \quad S(\omega) = \frac{D}{\pi} \frac{1}{\omega^2 + \alpha^2}. \tag{11}$$

(Pavliotis, Example 1.15) Circulant embedding runs the same Bochner/FFT machinery as an exact stationary sampler.

Proves Two independent estimators of S (an averaged Welch periodogram and the FFT of the estimated autocorrelation) match the Lorentzian (11) over the resolved band, and the total-variance normalization holds: $\int_{\mathbb{R}} S\, d\omega = D/\alpha = R(0)$ under the $1/2\pi$-on-S convention of (3). Circulant samples reproduce the analytic covariance to floating-point scale. The headline artifact is the estimated-vs-analytic density on log–log axes plus the pinned normalization scalar. *[External / not in Pavliotis: that the raw periodogram is a statistically* inconsistent *estimator of S and must be smoothed or averaged — a standard time-series fact (Percival & Walden; Brockwell & Davis).]*

Falsifier Two controls, both plotted. (i) The raw periodogram overlaid on Welch: it stays jagged and its variance does not shrink with record length at fixed ω, while Welch converges — inconsistency made visible. (ii) Dropping the $1/2\pi$ (engineering convention): the normalization ratio lands on exactly 2π instead of 1 — the spurious factor made to appear on purpose. (This README carries the 2π lesson for the whole repository.)

Depends on / enables $\Leftarrow$ Unit 0. Enables Units 3, 4, 6.

Deliverable shape `kernels.jl` (the exponential/OU kernel), `spectral.jl` (the Welch and raw periodogram estimators, the Bochner forward/inverse transforms, and the total-power and total-variance integrators), and `sampling.jl` (circulant embedding). The `experiments/01_spectral_bochner` experiment emits the density plot with the resolved band shaded, a raw-vs-Welch overlay, a circulant covariance-error panel, and the dropped-$1/2\pi$ control.

Effort & risk $\approx$ 8–9 h — the highest overrun risk in the project. Dominant risk: the FFT/normalization family (one- vs two-sided spectra, angular vs ordinary frequency, unnormalized FFTW, window power, circulant scaling) — pin the normalization once so the discrete $\int \widehat{S}\, d\omega \to R(0)$ and unit-test it.

Status Build: **shipped**. Cut: **core**.

Unit 2 — Karhunen–Loève via a Quadrature Eigenproblem (Torus Contrast + Truncation Cost)

Idea The KL expansion (Pavliotis, §1.5, Thm. 1.28) and Mercer's theorem give the operator's own adapted-basis diagonalization; Brownian motion has a closed-form KL basis on $[0, 1]$,

$$\lambda_n = \left(\frac{2}{(2n-1)\pi}\right)^2, \qquad e_n(t) = \sqrt{2}\,\sin\!\left(\tfrac{1}{2}(2n-1)\pi t\right), \tag{12}$$

(Pavliotis, Example 1.29, eqn. 1.37) solved numerically by a Nyström quadrature eigenproblem, *symmetrized* before the eigensolve.

Proves The numerical eigenpairs match the closed form (12) and the eigenvalue decay follows $\lambda_k \sim k^{-2}$; the trace identity $\sum_n \lambda_n = \int_0^T R(t,t)\,dt$ pins assembly (this equals $T^2/2$ for Brownian motion — *not* $T\,R(0) = 0$; the wrong target is itself worth recording), with the stationary specialization $\sum_n \lambda_n = T\,R(0)$ (Pavliotis, Exercise 23) checked on a stationary kernel; and the torus coincidence $\lambda_k = \hat{R}(k)$ holds *exactly* on the circle. The decay is a *deterministic* slope (its residuals are discretization misspecification, not sampling noise), gated only over the resolved range above the discretization floor — see conventions.

Falsifier Two, both latent in the design. (i) Run the torus coincidence on $[0,T]$: it *fails*, because the interval eigenfunctions are Sturm–Liouville sines, not characters. (ii) Extend the eigenvalue plot *below* the discretization floor: the k^{-2} slope flattens — the resolution limit made visible rather than hidden.

Depends on / enables $\Leftarrow$ Unit 0 (not Unit 1). Enables Units 3, 6.

Deliverable shape `kl.jl` (quadrature nodes and weights, the symmetrized Nyström eigensolver, the diagonal-trace quadrature, the KL tail-energy), `sampling.jl` (the KL truncation route), and `kernels.jl` (a torus kernel with a *full* positive spectrum — derived in-house, since Pavliotis's Exercise 27 verifies $\lambda_k = \hat{R}(k)$ only for the degenerate rank-2 kernel $\cos 2\pi(t-s)$, insufficient for the torus contrast wanted here). The `experiments/02_kl_quadrature` experiment emits the eigenfunctions, the eigenvalue decay with the resolved window shaded and the floor annotated, side-by-side torus-Fourier vs. interval Sturm–Liouville mode shapes, and the truncation-error-vs-K and KL-vs-Cholesky cost curves.

Effort & risk $\approx$ 10–11 h — the deepest unit. Dominant risk: the Nyström symmetrization (the naive discretization is non-symmetric even though R is) and the eigenvalue noise floor — both in the risk register.

Status Build: **shipped**. Cut: **core**.

Unit 3 — Reconciliation: The Process Zoo

Idea A small catalogue of processes — Brownian motion $R(t,s) = \min(t,s)$ (Pavliotis, eqn. 1.21), Brownian bridge $R(t,s) = \min(t,s) - ts$ (Pavliotis, eqn. 1.26 / Exercise 6), stationary OU $R(\tau) = \frac{D}{\alpha}e^{-\alpha|\tau|}$ (Pavliotis, Example 1.15) — each just a choice of kernel. The unit's one distinct idea is *reconciliation*: the sampling routes and the two diagonalizations are forced to agree on concrete processes, and *distributional* (not merely moment) identities are verified from finite samples. The sampling itself is reuse of Units 0/1/2.

Proves Three things. *(a) Route equivalence:* Cholesky, KL, and circulant samples of OU yield the same empirical covariance, judged against a *split-half null* — the same statistic computed between two halves of a single route's samples, which is by construction a draw from the "same law, different samples" distribution and so calibrates the gate without a variance formula. (This is essential because $\|\widehat{\Sigma}_A - \widehat{\Sigma}_B\|_F$ is a norm over many entries: it concentrates about a strictly positive value, not zero, so a "within k SE of zero" predicate gates against the wrong null and passes routes that disagree by a factor of two.) *(b) Distributional identities*, in two steps that are "the chain, not the conclusion": the transformed process is Gaussian *by construction* (a deterministic linear time-change of a Gaussian process is Gaussian, laws being closed under linear maps) — a proof, not something the samples establish; App. B.5 then upgrades a match of the *full* covariance into equality in law (e.g. $c^{-1/2}W(ct) \overset{d}{=} W(t)$); and Cramér–Wold projections — fixed linear functionals $a^\top X$, each KS-tested against $N(0, a^\top \Sigma a)$ — supply the evidence that would catch a non-Gaussian impostor carrying the right covariance, which is precisely the evidence App. B.5 alone cannot give. The KL coefficients are checked the same way (empirical correlation of the ξ_k is $\approx I$ and each ξ_k is KS-tested against $N(0, \lambda_k)$: uncorrelated *and* jointly Gaussian $\Rightarrow$ independent, the KL expansion's actual content). Bridge endpoints are pinned at $t = 0, 1$ via the covariance/Cholesky route, because the time-change formula $B_t = (1-t)W(t/(1-t))$ cannot be evaluated at $t = 1$. *(c) The cross-check:* on $[0, T]$ the OU KL eigenvalues converge, by the large-T Toeplitz/Grenander–Szegő limit, to $\lambda_k \to \hat{R}(\omega_k)$, $\omega_k = k\pi/T$, where $\hat{R}(\omega) = 2\pi\, S(\omega)$ is the *un-normalized* operator symbol — the Nyström eigenproblem carries no $1/2\pi$, so its eigenvalues converge to $\hat{R}$, not to the Unit-1 density S (which is 2π smaller). The convergence is distributional, not uniform in k (Szegő controls linear statistics, weakest at the spectral edges), so the gate is on the resolved-bulk gap shrinking with T — never a $\max_k$ over the whole spectrum — and no theoretical decay exponent is claimed. This is a *deterministic* slope (the gap is computed against the analytic $\hat{R}$, not a Welch estimate). *[External / not in Pavliotis: the Toeplitz/Szegő asymptotic, the Cramér–Wold device, and the Kolmogorov–Smirnov statistic and its finite-sample critical values are standard but external to Pavliotis.]*

Falsifier Repeat the self-similarity identity with the *wrong* exponent, $c^{-1/3}W(ct)$: the empirical covariance no longer matches $\min(t, s)$ (it is off by $c^{1/3}$), which is what makes the correct $c^{-1/2}$ meaningful. Fully self-contained.

Depends on / enables $\Leftarrow$ Units 0, 1, 2 (the three sampling routes). Enables Units 5 and 7.

Deliverable shape Reuses `sample_cholesky` / `sample_kl` / `sample_circulant_embedding`, `empirical_cov` (also per half, for the split-half null), `nystrom_eigen` (the cross-check eigenvalues), the three catalogue kernels including a new Brownian-bridge kernel, and Welch (only for the visual S-overlay in the artifact, never the gate). It introduces *one* new `src/` module, `gof.jl` (a Kolmogorov–Smirnov statistic): deliberately *not* an operation on $\mathcal{C}$ — the one considered exception to the organizing principle — placed in `src/` rather than the experiment folder because two units need it (Cramér–Wold here, Donsker rates in Unit 5) and `src/` is the tier that gets a deterministic test. The `experiments/03_process_zoo` experiment emits a three-panel portrait per process, a route-equivalence panel plotting the pairwise distances against the split-half null band, a Cramér–Wold per-projection panel, the KL-coefficient correlation/normality check, the pinned bridge endpoints, the cross-check $\log g(T)$-vs-$\log T$ plot with its fitted slope, and the wrong-exponent control.

Effort & risk $\approx 8\,\text{h}$ — almost entirely reuse; the added hour is the Cramér–Wold projections, the split-half calibration, and `gof.jl` with its deterministic test. Dominant risk: the estimator-noise-floor family applied to the cross-check bulk gap — gate on the bulk, never on a $\max_k$ where the edge eigenvalues Szegő does not control can stall the decay.

Status Build: **shipped**. Cut: **core**.

Unit 4 — Ergodicity as Loop-Closer

Idea The ergodic properties of stationary processes (Pavliotis, §1.2): the L^2 law of large numbers (Pavliotis, Prop. 1.16, (8)), valid under $C \in L^1(0,\infty)$; the finite-T variance identity (Pavliotis, Lemma 1.17, (9)); and the Green–Kubo formula (10).

Proves For stationary OU, the variance of the time-average obeys $\mathrm{Var}\big(\frac{1}{T}\int_0^T X_s\,ds\big) \to \frac{2D_\star}{T} = \frac{2D}{T\alpha^2}$ — both the slope -1 and the constant, the latter carrying α — and the integrated process grows as $\mathbb{E}\big(\int_0^t X_s\,ds\big)^2 \to 2D_\star t = \frac{2D}{\alpha^2}t$. The headline artifact is the variance-vs-T log–log slope gated stochastically against -1, together with the fitted constant against $2D_\star$ (the constant is what pins α). *This is the methodological keystone*: it is the justification for the whole project's one-path-vs-analytic methodology used in Units 0–3.

Falsifier Feed the *same* time-average-variance estimator a synthetic stationary process with a *non-integrable* correlation, $C(u) \propto (1+u)^{-1/2}$ (so $C \notin L^1$): the variance decays slower than $1/T$ and the fitted slope departs from -1 — Prop. 1.16's L^1 hypothesis shown load-bearing. Self-contained, with no forward reference to fBm (Unit 6 runs the genuine-process version).

Depends on / enables $\Leftarrow$ Units 0 and 1 — reuses Unit-1 circulant (preferred) or Unit-0 Cholesky for OU paths. Enables Unit 6.

Deliverable shape A new `ergodic.jl` (a running time-average along one path, the time-average variance across an *ensemble* of length-T paths, the integrated mean-square displacement, and the Green–Kubo coefficient $D_\star = D/\alpha^2$), plus the reused OU sampler. The `experiments/04_ergodicity` experiment emits the variance-vs-T log–log with the fitted $1/T$ slope and constant, the MSD-vs-t with fitted slope, the running-average path with a $\pm\sqrt{2D_\star/T}$ band, and the non-integrable-C control.

Effort & risk $\approx 6\,$h. Dominant risk: confusing the time-average with its variance — one long path gives a single realization, not the variance, which needs an ensemble (a path *matrix*), so compute the $\pm\sqrt{2D_\star/T}$ band with $D_\star = D/\alpha^2$, not D.

Status Build: **planned**. Cut: **core**.

Unit 5 — Brownian Motion as a Scaling Limit (Random Walk → BM)

Idea Brownian motion as the weak (Donsker) limit of a rescaled random walk (Pavliotis, §1.3): for iid mean-zero, variance-one increments the interpolated, rescaled walk converges weakly to a standard Brownian motion. This is BM's *construction* — a statement about laws on path space, complementary to the covariance-operator picture and deliberately off the $\mathcal{C}$-spine.

Proves The finite-dimensional distributions and a path functional (the running maximum) converge to the Gaussian/BM target: the one-time marginal $\Rightarrow N(0,1)$, the two-time covariance $\to \min(s,t)$. The *marginal-convergence rate is law-dependent*: Berry–Esseen bounds the Kolmogorov–Smirnov statistic by $O(n^{-1/2})$, but that rate is achieved only when the skewness is nonzero — for a *smooth* symmetric increment the leading $n^{-1/2}$ Edgeworth term vanishes, leaving the $O(n^{-1})$ kurtosis term. A *lattice* symmetric law, however, carries a separate, skewness-independent Esseen lattice-discreteness term that is itself $O(n^{-1/2})$ and does not vanish. So the target is $\propto n^{-1/2}$ for the *exponential* (skewed) increment *and* for *Rademacher* (symmetric but lattice, via discreteness), and $\propto n^{-1}$ only for the smooth-symmetric *uniform* increment — three mechanisms, two rates. The headline artifact is the KS-vs-n log–log plot with one line per law, each slope gated stochastically. *[External / not in Pavliotis: Donsker's invariance principle / the functional CLT; Berry–Esseen for the $n^{-1/2}$ bound and the Edgeworth expansion for the n^{-1} symmetric-law rate. Standard, external to the second-order framing; high confidence.]*

Falsifier Add an *infinite-variance* increment (symmetric Pareto, tail index < 2): the rescaled walk does not converge to $N(0,1)$ at all — the marginal departs from the QQ line, its empirical variance *grows* with n instead of settling at 1, and $n^{-1/2}$ is simply the wrong normalization — showing Donsker's finite-variance hypothesis is load-bearing.

Depends on / enables $\Leftarrow$ Units 0 (the empirical covariance) and 3 (`gof.jl`'s KS statistic). Enables nothing.

Deliverable shape *No new `src/` code* — the rescaled-walk builders live in the experiment, weak convergence of laws not being an operation on $\mathcal{C}$; the one library call is `gof.jl`'s `ks_statistic`, introduced by Unit 3 and reused rather than duplicated. The `experiments/05_bm_scaling_limit` experiment emits overlaid rescaled paths at increasing n, a marginal histogram/QQ against $N(0,1)$, the KS-vs-n log–log with one line per increment law, the empirical-vs-$\min(s,t)$ covariance, the running-maximum across laws, and the infinite-variance control.

Effort & risk $\approx$ 4–5 h (plus a small buffer for the law-dependent-rate surprise). Dominant risk: expecting $n^{-1/2}$ for the symmetric laws — a "too steep" -1 slope is correct, not a bug.

Status Build: **planned**. Cut: **core**.

Unit 6 — Fractional Brownian Motion

Idea A one-exponent generalization of Brownian motion tuning both self-similarity and increment correlation (Pavliotis, Def. 1.26, Prop. 1.27), with covariance

$$R_H(t,s) = \tfrac{1}{2}\left(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\right). \tag{13}$$

(Pavliotis, eqn. 1.27)

Proves The empirical covariance matches R_H for several H; the self-similarity identity $W^H_{ct} \overset{d}{=} c^H W^H_t$ (Pavliotis, eqn. 1.28) holds as a covariance identity; and the eigenvalue decay of $\mathcal{C}_H$ contrasts with Brownian motion's k^{-2}. The headline artifacts are the covariance and self-similarity matches together with the eigenvalue-decay overlay — and, as the control below, the ergodic breakage at $H > \frac{1}{2}$.

Falsifier The full fBm version of Unit 4's control: apply the time-average-variance estimator at $H = 0.8$. The increments have non-summable correlations ($C \notin L^1$), so the $1/T$ ergodic rate breaks — the first genuine process where Unit 4's machinery fails, run rather than asserted.

Depends on / enables ⇐ Units 1 (circulant sampler, with the fBm minimal-nonnegative-embedding caveat), 2 (`nystrom_eigen` for the decay contrast), and 4 (the time-average-variance estimator for the breakage control); also Unit 0's Cholesky. Enables nothing.

Deliverable shape `kernels.jl` gains the fBm kernel; the unit reuses the circulant and Cholesky samplers, `nystrom_eigen`, and `time_average_variance`. The `experiments/06_fbm` experiment emits paths at $H \in \{0.3, 0.5, 0.8\}$ (rougher to smoother), covariance heatmaps, the self-similarity scaling check, the $\mathcal{C}_H$ eigenvalue-decay overlay against BM, and the $H = 0.8$ ergodic-breakage panel.

Effort & risk $\approx$ 5–6 h. Dominant risk: circulant non-negativity fails for fBm — the plain symmetric embedding can have negative eigenvalues, so enlarge to the minimal-nonnegative (Dietrich–Newsam) embedding *[External / not in Pavliotis: fBm circulant embedding (Dietrich–Newsam) and the nonnegative-embedding caveat are standard but external to Pavliotis.]* — part of the FFT/normalization and noise-floor families.

Status Build: **planned**. Cut: **cuttable — trim second**.

Unit 7 — SDE Bridge: OU Invariant Law and Strong vs. Weak Convergence

Idea The dynamical (SDE) view that bridges to Chapter 3. The OU SDE with *additive* noise,

$$dX_t = -\alpha X_t\, dt + \sqrt{2D}\, dW_t, \tag{14}$$

has stationary law $N(0, D/\alpha)$ with covariance $\frac{D}{\alpha}e^{-\alpha|\tau|}$ (Pavliotis, cf. Ch. 2, Example 2.5) — the $\sqrt{2D}$ makes the stationary variance agree with the Unit-3 kernel — and its Euler–Maruyama discretization. The strong/weak-convergence *distinction* is demonstrated on geometric Brownian motion with *multiplicative* noise,

$$dY_t = \mu Y_t\, dt + \sigma Y_t\, dW_t, \qquad Y_t = Y_0 \exp\big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\big), \tag{15}$$

which has a closed-form solution against which both errors can be measured.

Proves The EM OU invariant law converges to $N(0, D/\alpha)$ with covariance $\to \frac{D}{\alpha}e^{-\alpha|\tau|}$ (checked against Unit 3's exact stationary OU); and for GBM the strong error $\mathbb{E}|Y_T^{\Delta t} - Y_T| \propto \Delta t^{1/2}$ (slope $\frac{1}{2}$) while the weak error $|\mathbb{E}f(Y_T^{\Delta t}) - \mathbb{E}f(Y_T)| \propto \Delta t$ (slope 1). The headline artifact is the log–log plot of both GBM errors with their slopes gated stochastically *and* shown visibly distinct — the 0.5/1.0 gap being a multiplicative-noise phenomenon invisible to deterministic Taylor analysis. *[External / not in Pavliotis: the OU/GBM SDEs and the Euler–Maruyama/Milstein strong and weak convergence orders are Chapter-3 / standard-SDE-numerics material (Kloeden & Platen); used here only as an explicitly labelled forward pointer.]*

Falsifier The most striking falsifier in the project: recompute the GBM strong error with *independent* coarse and fine Brownian paths instead of the shared one — the "strong error" stops decaying and plateaus at an $O(1)$ constant, showing strong convergence is a statement about a *shared* noise realization. A near-free contrast reinforces it: the same integrator on OU (additive noise) shows strong order 1.0 — no gap — because the Milstein correction vanishes for additive noise, which is precisely *why* the strong/weak gap must be exhibited on multiplicative noise; on GBM the same integrator shows 0.5.

Depends on / enables ⇐ Unit 3 (the exact stationary OU the invariant-law check is compared against). Enables nothing — it is the forward pointer.

Deliverable shape A new `sde.jl` (an Euler–Maruyama step, an additive-noise OU step, a GBM closed-form reference, a path simulator from a step rule, a driving Brownian path with its increments, and a coarsen-increments helper that shares one driving path across resolutions). The `experiments/07_sde_bridge` experiment emits the OU invariant-law check (histogram/QQ and covariance against the exact kernel), the GBM strong (shared path) and weak (many paths) errors log–log with both slopes, the un-coupled strong-error plateau, and the OU-vs-GBM strong-order contrast.

Effort & risk ≈ 8 h. Dominant risk: the coupling trap (strong-error checks require the *same* driving path at coarse and fine steps — generate the fine path and subsample it; the un-coupled version is the control, not the method) and the weak-error Monte-Carlo floor — both in the risk register.

Status Build: **planned**. Cut: **cut line — trim first** (a pure Chapter-3 forward pointer; nothing in Units 0–6 depends on it).

4 Repository Architecture and Rationale

The repository separates a small, tested **library** (`src/`, organized by *operation on* $\mathcal{C}$) from a narrative, reproducible **experiment gallery** (`experiments/`, organized by *unit*). This separation is the central design decision. Units 00–03 are shipped; 04–07 are planned and are *not* pre-stubbed.

```
StochasticProcesses.jl/
|-- Project.toml, Manifest.toml     # both committed: exact reproducibility
|-- README.md                       # through-line + gallery index
|-- src/
|   |-- StochasticProcesses.jl      # module definition, flat re-exports
|   |-- kernels.jl                  # R(t,s): min, min-ts, exp/OU, periodic; (+fBm, Unit 6)
|   |-- gaussianprocess.jl          # GP type, grid assembly, empirical covariance
|   |-- sampling.jl                 # the square roots: Cholesky / KL / circulant (FFTW)
|   |-- spectral.jl                 # Welch & raw periodogram, Bochner transforms, integrators
|   |-- kl.jl                       # Nystrom quadrature eigenproblem (symmetrized)
|   |-- gof.jl                      # KS statistic; the considered exception -- see below
|   |-- ergodic.jl                  # time-average estimators, Green-Kubo   (Unit 4, planned)
|   `-- sde.jl                      # Euler-Maruyama, coupling helpers      (Unit 7, planned)
|-- test/                           # analytic identities, tight tolerances (CI runs this)
|   `-- runtests.jl
|-- experiments/                    # the gallery -- one folder per unit
|   |-- 00_covariance_core/     {run.jl, README.md, figures/}   # shipped
|   |-- 01_spectral_bochner/    {run.jl, README.md, figures/}   # shipped
|   |-- 02_kl_quadrature/       {run.jl, README.md, figures/}   # shipped
|   |-- 03_process_zoo/         {run.jl, README.md, figures/}   # shipped
|   |-- 04_ergodicity/          ...                             # planned
|   |-- 05_bm_scaling_limit/    ...                             # planned
|   |-- 06_fbm/                 ...                             # planned
|   `-- 07_sde_bridge/          ...                             # planned (Ch.3 pointer)
```

```
|-- experiments/Project.toml, Manifest.toml   # separate env that `dev`s the package
`-- .github/workflows/ci.yml
```

Rationale. `src/` is organized by *operation on* $\mathcal{C}$ — kernels, the two diagonalizations, the square roots, the ergodic loop — rather than by named process (Brownian motion, OU, . . .). A process is nothing more than a choice of kernel; organizing by operation makes the module structure a faithful expression of the through-line *for the second-order theory of Chapter 1.* Two units sit deliberately at the edge of that spine and are placed among `experiments/` accordingly: Unit 5 (random walk $\to$ BM) is weak convergence of *laws*, not an operation on $\mathcal{C}$, and its builders live entirely in its experiment folder; Unit 7 (SDE bridge) is a dynamical, generator-flavored Chapter-3 pointer whose only library home is `sde.jl`.

One module, `gof.jl`, is a *considered exception* to the organizing principle: a Kolmogorov–Smirnov statistic is not an operation on $\mathcal{C}$ by any reading, but Units 3 and 5 both need one — for Cramér–Wold projections and for Donsker rates respectively — and the alternative is the same few lines duplicated across two experiment folders where the testing convention gives them no deterministic test at all. It is placed in `src/` for the test coverage and the single definition, and the principle is stated as *bending* here rather than quietly redefined to accommodate it.

The rationale's limit, stated with the rationale. The covariance-operator spine is the right organizing principle precisely because Chapter 1 is the Gaussian/second-order chapter. Adding later-chapter material never forces a rewrite of existing files — one adds modules to `src/` and numbered folders to `experiments/` — but the covariance operator ceases to be the hub. Chapter 2 (diffusions, Markov generators, the Kolmogorov equations) and Chapter 4 (Fokker–Planck) are centered on the *generator*/transition semigroup on generally non-Gaussian processes; they introduce a second, generator-centered spine *alongside* $\mathcal{C}$, rather than more operations on it. So the code accretes without restructuring, but the single-hub *narrative* is Chapter-1-specific and should not be over-sold as the framework's permanent organizing idea. Each experiment folder remains self-contained and documents concept $\to$ implementation $\to$ theoretical check $\to$ result, which directly serves the goal of leaving durable evidence of understanding.

5 Cross-Cutting Conventions

These are load-bearing for reproducibility; violating one silently corrupts every downstream number. The concrete values (seeds, jitter, grid sizes, tolerances) are recorded in each experiment, not here.

- **Seeding.** Use `StableRNGs.jl`, never Julia's default global RNG. *[External / not in Pavliotis: the default stream is not guaranteed stable across Julia versions, so seeded Monte Carlo checks can silently drift; a Julia-ecosystem fact, not a Pavliotis one.]* Pass an explicit `StableRNG(seed)` to every stochastic routine and record the seed. Do not reorder RNG draws within an experiment — demo paths and the ladder pull from one stream, so reordering changes every committed number.
- **Numerical positive-definiteness (Cholesky).** Form Σ on the grid and factor $\Sigma + \varepsilon I$ with a small jitter (nugget) ε, and *report* it: too small gives an indefinite throw, too large biases the covariance. Prefer the KL or circulant route for the smoothest kernels where even jittered Cholesky loses accuracy.
- **Spectral normalization.** Pin the periodogram and Bochner-FFT normalizations so the discrete $\int \hat{S}\, d\omega \to R(0)$, matching the $1/2\pi$-on-S convention of (3); a mismatched normal-

ization is the most likely source of a spurious factor of 2π. Document the chosen normalization (and the circulant-embedding scaling) in the relevant experiment README.

- **Two verification tiers.**

  1. `test/`: deterministic analytic identities — kernel symmetry/PSD, the trace identity in its general form $\sum_n \lambda_n = \int_0^T R(t,t)\,dt$ (plus the stationary specialization $T\,R(0)$), Brownian-motion eigenvalues vs. the closed form (12) — with tight absolute tolerance, over the resolved eigenvalue range only. This is what CI runs.

  2. `experiments/`: Monte Carlo checks with tolerance a small multiple of the estimated standard error, seeded so results are deterministic given the seed.

- **Tests must bite.** Prefer hand-computed targets over re-running the function's own formula, and non-square / asymmetric fixtures so shape, orientation, and one-sided-implementation bugs cannot hide (e.g. the KS statistic is exercised on skewed fixtures, since a symmetric case does not discriminate a one-sided implementation).

- **Rate checks as the primary evidence.** A measured *slope* on a log–log plot is the headline artifact for every unit, and is stronger evidence of correct implementation than a single-point match. The two classes of slope are gated differently, and *the distinction is not whether the process is random but whether the regression residuals are sampling noise*:

  – *Stochastic slopes* (the Monte-Carlo covariance error $\propto N^{-1/2}$; the Euler–Maruyama strong/weak orders on GBM; the ergodic variance $\propto 1/T$; the Donsker KS-vs-n rates of Unit 5): fit on log–log axes and gate against the theoretical value to within a small multiple of the fitted slope's own standard error — the estimator variance differs sharply between checks, so a single fixed absolute tolerance would be simultaneously too loose in one place and too tight in another.

  – *Deterministic slopes* (the KL eigenvalue-decay slope of Unit 2; the Toeplitz bulk-gap slope of Unit 3): the regression residuals are model misspecification — the asymptotic k^{-2} plus the discretization floor for Unit 2, the Szegő asymptotic plus quadrature error for Unit 3 — *not* sampling noise, so a regression standard error is not a sampling standard error and an SE-based gate is not meaningful. Fit only within the resolved window and gate with a fixed absolute margin: for Unit 2 against the theoretical exponent -2; for Unit 3, where no defensible theoretical exponent exists, against a fixed negative threshold, reporting the fitted value rather than claiming it. Unit 3's gap is deterministic precisely because it is computed against the analytic symbol rather than a Welch estimate.

- **Continuous integration.** Run `test/runtests.jl` on push via GitHub Actions — fast, guarding the deterministic analytic core. Do *not* run the slower Monte Carlo experiments in CI; run those locally and commit the resulting figures directly. Experiments are headless (`ENV["GKSwstype"]="100"` before `gr()`).

- **Environment pinning.** Commit `Manifest.toml` alongside `Project.toml` (for *both* the root package env and the separate `experiments/` env, which `dev`s the package via a relative path) so `Pkg.instantiate()` reproduces exact versions. Manifest pinning plus explicit RNG seeding gives full end-to-end reproducibility of every figure.

6 Consolidated Risk Register

Three failure families recur across units; internalize them once rather than per unit.

- *The FFT/normalization family* (Units 1, 6, 7): one- vs two-sided spectra; angular (ω) vs ordinary ($f = \omega/2\pi$) frequency, worth a factor 2π; FFTW returns unnormalized transforms (restore Δt, $1/N$); Welch divides by the window power $\sum_n w_n^2$, not N; the circulant-embedding scaling is validated once, in Unit 1. Single guard: pin every normalization so the discrete $\int \widehat{S}\, d\omega \to R(0)$ and unit-test it.
- *The estimator noise floor* (Units 2, 7, and the raw periodogram of Unit 1): every rate check has a level below which the estimator's own variance or discretization dominates and the slope flattens — the Nyström eigenvalue floor, the weak-error Monte-Carlo floor, and the inconsistent raw periodogram are the *same* phenomenon. Single guard: identify the floor, fit slopes only above it, and state the resolved range.
- *The strong-convergence coupling* (Unit 7): strong-error checks require the *same* driving Brownian path at coarse and fine steps — generate the fine path first and subsample it. The un-coupled version is a negative control, not the method.

7 Budget

At the working budget of 15 h/week (6 × 2.5 h) against a 6–8 week horizon, i.e. 90–120 h, the summed per-unit estimates total ≈ 55–61 core hours; the remainder is deliberate headroom for per-unit spec-writing and review of all generated code, plus a debugging buffer concentrated on the units most likely to overrun for someone newer to stochastic numerics: **Units 1, 2, 6, and 7**. The risk in Units 1, 6, and 7 is spectral/FFT normalization; the risk in Unit 2 is the Nyström symmetrization and the eigenvalue noise floor. One further hidden time-sink: the Unit-5 marginal-convergence rate is law-dependent, so a fitted slope that is not $n^{-1/2}$ can be correct rather than a bug.

Reconciled against shipped reality: **Units 0–3 are landed**, accounting for roughly 32–36 of the core hours; the **remaining** committed work (Units 4–7) is roughly 23–25 hours, so the split reconciles with the total rather than merely relabeling it. Everything through Unit 6 is firm; the SDE bridge (Unit 7) is the single cut line, with Unit 6 trimmed second, if the buffer is ever consumed — nothing in Units 0–6 depends on Unit 7. This spends the extra budget over a leaner plan on *more of Chapter 1 done thoroughly* (the scaling-limit construction, fBm, the truncation-cost study, circulant embedding), consistent with the stated priority of depth over breadth.

Sources

Call signatures, implementations, and numerical tolerances are *not* recorded in this plan: the shipped code under `src/` is the source of truth for the former, and each feature brief is handed to `plan-and-dispatch`, whose per-commit plans own the contracts and the gate values. (The previous revision of this document carried an appendix of illustrative code stubs and per-unit signature tables; those were deleted as downstream content that competes with the real source of truth.)

- G. A. Pavliotis, *Stochastic Processes and Applications*. The project's copy is the *November*

11, 2015 draft; all section, theorem, and equation numbers cited here (Chapter 1, plus §B.5 and a single Chapter-2 pointer, Example 2.5) refer to that draft and were checked directly against it. Its numbering differs from the published Springer edition. In particular, **Mercer's theorem is stated inline and unnumbered in §1.5** (there is no "Theorem 1.18"; "Example 1.18" is the exponential-covariance / diffusion-coefficient example, where $\int_0^\infty C = D/\alpha^2$ is computed explicitly), and the ergodic result appears as *Proposition 1.16* (its proof header inconsistently reads "Theorem 1.16").

- External references, flagged inline where used and not verified against Pavliotis: Grenander & Szegő, *Toeplitz Forms and Their Applications* (asymptotic KL/spectral-density coincidence off the torus); D. B. Percival & A. T. Walden, *Spectral Analysis for Physical Applications*, or P. J. Brockwell & R. A. Davis, *Time Series: Theory and Methods* (periodogram inconsistency); C. R. Dietrich & G. N. Newsam, and A. T. A. Wood & G. Chan (circulant embedding of stationary Gaussian fields); P. E. Kloeden & E. Platen, *Numerical Solution of Stochastic Differential Equations* (Euler–Maruyama/Milstein strong and weak orders, including strong order 1 for additive noise); W. Feller, *An Introduction to Probability Theory and Its Applications, Vol. II* (the Edgeworth expansion / skewness-dependent Berry–Esseen rate of Unit 5); the functional central limit theorem / Donsker's invariance principle (Unit 5); and P. Billingsley, *Probability and Measure* (the Cramér–Wold device) together with any standard nonparametric-statistics reference for the Kolmogorov–Smirnov statistic and its finite-sample critical values (both used for the Unit-3 distributional identities).